

Integer Arithmetic



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Relative Prime Integers

Modular Arithmetic

Set of Residues

Congruence

Residue Classes

Operations in Z_n

Relatively Prime Integers

Two integers a and b are said to be relative prime or co-prime if $\gcd(a,b)=1$

e.g Two numbers 16 and 21 are relatively prime as $\gcd(16,21)=1$

1. 15,28 are relatively prime?
2. 32,63 are relatively prime?

Note: If a and b are relative primes then there exists integer x and y such that $ax+by=1$ (gcd of a and b).

As an example, the greatest common divisor of 5 and 3 is 1, and we can write $5*(2)+3*(-3)=1$.

$$a \bmod n = r$$

here a is any integer $\mathbb{Z}$, n should be a positive integer and r should be non-negative.

n is called "modulus" and r is called the "residue".

Examples

A) $27 \bmod 5 = 2$

B) $36 \bmod 12 = 0$

C) $-18 \bmod 14 = -4$

Here r is negative. So to make it non-negative, **add the modulus**.

$$-4 + 14 = 10$$

therefore $r = 10$

D) $-7 \bmod 10 = -7 = -7 + 10 = 3$

Set of Residues : Z_n

The result of the modulo operation with modulus n is always an integer between 0 and $n-1$.

That is, the result is always a non-negative integer less than n .

Therefore, the modulus operation creates a set called the set of least residues modulo n or Z_n .

$$Z_n = \{0, 1, 2, 3, \dots, (n-1)\}$$

e.g. $Z_2 = \{0, 1\}$

$$Z_6 = \{0, 1, 2, 3, 4, 5\}$$

$$Z_{11} = \{0, 1, 2, \dots, 10\}$$

Let a and b be integers. Then $a \equiv b \pmod{m}$ is read as " a is congruent to b modulo m "

This means that they both leave the same remainder when divided by m .

e.g. $2 \equiv 12 \pmod{10}$

$-8 \equiv 2 \pmod{5}$

$Z_{10} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

$-8 \equiv 2 \equiv 12 \equiv 22 \pmod{10}$

Let a, b, c and d be integers. Then following are the properties of Congruence Relation.

1. If $a \equiv b \pmod{m}$, then $b \equiv a \pmod{m}$
2. If $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$, then $a \equiv c \pmod{m}$
3. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then $a + c \equiv b + d \pmod{m}$
and $a - c \equiv b - d \pmod{m}$
4. If $a \equiv b \pmod{m}$, then $ac \equiv bc \pmod{m}$
5. If $a \equiv b \pmod{m}$, then $a^k \equiv b^k \pmod{m}$, for all $k \geq 1$

Let m be a positive integer and a be any integer, then

$$[a]_m = \{x : x \equiv a \pmod{m}\}$$

e.g. if $m=5$, we have 5 sets

$$[0], [1], [2], [3], [4]$$

$$[0] = \{\dots, -15, -10, 0, 5, 10, 15, \dots\}$$

$$[1] = \{\dots, -14, -9, -4, 1, 6, 11, 16, \dots\}$$

$$[2] = \{\dots, -13, -8, -3, 2, 7, 12, 17, \dots\}$$

$$[3] = \{\dots, -12, -7, -2, 3, 8, 13, 18, \dots\}$$

$$[4] = \{\dots, -11, -6, -1, 4, 9, 14, 19, \dots\}$$

- A) $(a+b) \bmod n$
- B) $(a-b) \bmod n$
- C) $(a \times b) \bmod n$

e.g.

1. Add 7 to 14 in Z_{15}
2. Subtract 11 from 7 in Z_{13}
3. Multiply 11 by 7 in Z_{20}

1. $(a+b) \bmod n = [(a \bmod n) + (b \bmod n)] \bmod n$
2. $(a-b) \bmod n = [(a \bmod n) - (b \bmod n)] \bmod n$
3. $(a \times b) \bmod n = [(a \bmod n) \times (b \bmod n)] \bmod n$